Band-1 Hadamard Order 8564 – Miyamoto (Paley-II Seed H_2140 q=1069 + C_2142) + Paley-I q=8563 IN SPK; Gap 2052 Still OPEN

H_2140 Paley-II PASS; C_2142 PASS; H_8564 Miyamoto PASS; H_8564 Paley-I q=8563 PASS -> IN SPK via Paley-I; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8516_Miyamoto`; `2026-09-16_Band1_Hadamard_8356_PaleyII`.
Lab. `/workspace/experiments/band1/hadamard_8564_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2140 and 8564 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{2140} = Paley-II($q = 1069$) and symmetric Paley conference C_{2142} over GF(2141) ($q = 2141 \equiv 1 \pmod 4$), **and** classical Paley-I H_{8564} = Paley-I($q = 8563$) (prime $\equiv 3 \pmod 4$). Classical SPK for 8564: Paley-I **PASS**es ($q = 8563$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL**s ($q = 4281 = 3 \cdot 1427$ NOT prime power); Kronecker **FAIL**s (4×2141). Therefore order **8564 is IN SPK via Paley-I** – Miyamoto alone never folds an order into SPK. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8516 Miyamoto-only NOT in SPK** (seeds **1064+2128**). Prior SPK seeds **1064 / 2128 / 2112 / 2088+8356 / 2028+8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2140** and **8564** folded into ONBOX/SPK when verified (8564 via Paley-I only).

Abstract

Finite constructions of orders **2140** and **8564**.

- H_{2140} = Paley-II ($q = 1069$) – preferred seed (Paley-I $q = 2139$ FAIL composite; Double FAIL; Kronecker 4×535 FAIL)
- C_{2142} = Paley conference over GF(2141), $q = 2141$
- H_{8564} = Miyamoto(H_{2140}, C_{2142})
- H_{8564} = Paley-I ($q = 8563$) – classical SPK path
- Paley-II for 8564: **SKIP** ($q = 4281 = 3 \cdot 1427$ NOT prime power)
- Kronecker for 8564: **SKIP** ($4 \cdot 2141$ not admissible)

SPK: 8564 IN SPK via Paley-I $q = 8563$. Seeds 2140 and 8564 enter SPK. Prior SPK **1064 / 2128 / 2112 / 2088 / 8356 / 2028 / 8116 / 1996 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8564.py`

Results. overall_pass=true; units 154/154; wall 1060.4434s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8564 = 4 \cdot 2141$; 2141 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2140} and C_{2142} (Paley conference over prime field $q = 2141$).
- Seed: Paley-II $q = 1069$ (prime $\equiv 1 \pmod 4$) -> H_{2140} (preferred); Paley-I $q = 2139$ FAIL; Double FAIL; Kronecker 4×535 FAIL.
- SPK: Paley-I $q = 8563$ **PASS** (prime $\equiv 3 \pmod 4$); Paley-II $q = 4281$ **FAIL** (NOT prime power $3 \cdot 1427$); Kronecker **FAIL** -> **IN SPK via Paley-I**.
- Prior: 8516 Miyamoto-only NOT in SPK shipped; this ship $q = 2141 \rightarrow n = 8564$ (+ Paley-I SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_2140 Paley-II	Paley-II q=1069 DIRECT	0	5.1869
C_2142	Paley conf GF(2141) q=2141	0	6.3722
H_8564 Miyamoto	Miyamoto(H_2140 Paley-II, C_2142)	0	498.1544
H_8564 Paley-I	Paley-I q=8563 (classical SPK)	0	538.9844
H_8564 Paley-II	SKIP q=4281=3*1427 NOT prime power	–	–
H_8564 Kronecker	SKIP 4*2141 not admissible	–	–

SPK closure note

Seeds include prior SPK ships through **8356/8116** lineage (**1064** Paley-I; **2128** Double; **2112** Paley-I; **2088** Paley-I; **8356** Paley-II; **2028** Paley-I; **8116** Paley-II; **1996** Paley-II; **1932** Paley-I; **7604** Paley-I; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 8516/8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196 + verified **2140** (Paley-II) + verified **8564** (Paley-I q=8563); **exclude** Miyamoto-only bonuses. Known count 1668. Next constructive gap **2052** (OPEN).

Quantity	Value
8564 in SPK	**True** (Paley-I q=8563)
2140 in SPK	**True** (Paley-II q=1069)
1064 / 2128 / 2112 / 2088 / 8356 / 2028 / 8116 / 1996 in SPK	**True** (prior)
8516 / 8452 / 7988 / 7732 / 7492 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture * no Clay-closed * no finished-ToE * no RH / P!=NP / YM / Frankl / NS-solved / AC-solved * gap 2052 remains OPEN * finite constructions of 2140+8564 ONLY * Miyamoto alone never folds into SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_8564_miyamoto/`
Artifacts: `H2140_paleyII_pm1.npz`, `C2142_pm1.npz`, `H8564_miyamoto_pm1.npz`, `H8564_paleyI_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.